

The Exponent Rules (Part II)

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Two Rule Problems

Simplify each expression below and write the Roman numerals of the exponent rules that you used in the margin. Do not leave your answers with fraction exponents or negative exponents.

Problem Set A

I, VII ① $x^{-3} \cdot x^3 = x^{-3+3} = x^0 = \boxed{1}$

III, V ② $(4x^5)^2 = 4^2 \cdot (x^5)^2 = 16 \cdot x^{5 \cdot 2} = 16 \cdot x^{10} = \boxed{16x^{10}}$

II, VI ③ $\frac{x^6}{x^5} = x^{6-5} = x^1 = \boxed{x}$

VIII, IX ④ $(-32)^{-\frac{1}{5}} = \frac{1}{(-32)^{\frac{1}{5}}} = \frac{1}{\sqrt[5]{-32}} = \frac{1}{-2} = \boxed{-\frac{1}{2}}$

III, V ⑤ $(-2y^3)^4 = (-2)^4 \cdot (y^3)^4 = 16 \cdot y^{3 \cdot 4} = \boxed{16y^{12}}$

VIII, IX ⑥ $8^{-\frac{1}{3}} = \frac{1}{8^{\frac{1}{3}}} = \frac{1}{\sqrt[3]{8}} = \boxed{\frac{1}{2}}$

II, VI ⑦ $\frac{b^7}{b^6} = b^{7-6} = b^1 = \boxed{b}$

I, VII ⑧ $y^9 \cdot y^{-9} = y^{9+(-9)} = y^0 = \boxed{1}$

Problem Set B

$$\text{IV, V} \quad ⑨ \left(\frac{x^2}{y^4} \right)^3 = \frac{(x^2)^3}{(y^4)^3} = \frac{x^{2 \cdot 3}}{y^{4 \cdot 3}} = \boxed{\frac{x^6}{y^{12}}}$$

$$\text{V, VI} \quad ⑩ (x^9)^{\frac{1}{9}} = x^{9 \cdot \frac{1}{9}} = x^1 = \boxed{x}$$

$$\text{II, VII} \quad ⑪ \frac{5^{-3}}{5^{-3}} = 5^{-3 - (-3)} = 5^{-3 + 3} = 5^0 = \boxed{1}$$

$$\text{I, VIII} \quad ⑫ a^5 \cdot a^{-7} = a^{5 + (-7)} = a^{-2} = \boxed{\frac{1}{a^2}}$$

$$\text{V, VI} \quad ⑬ (5^{\frac{1}{2}})^2 = 5^{\frac{1}{2} \cdot 2} = 5^1 = \boxed{5}$$

$$\text{I, VIII} \quad ⑭ 10^{-7} \cdot 10^3 = 10^{-7 + 3} = 10^{-4} = \frac{1}{10^4} = \boxed{\frac{1}{10,000}}$$

$$\text{IV, V} \quad ⑮ \left(\frac{a^3}{b^4} \right)^2 = \frac{(a^3)^2}{(b^4)^2} = \frac{a^{3 \cdot 2}}{b^{4 \cdot 2}} = \boxed{\frac{a^6}{b^8}}$$

$$\text{II, VII} \quad ⑯ \frac{y^{-2}}{y^{-2}} = y^{-2 - (-2)} = y^{-2 + 2} = y^0 = \boxed{1}$$

Problem Set C

$$\text{V, X} \quad ⑰ (27^2)^{\frac{1}{3}} = 27^{2 \cdot \frac{1}{3}} = 27^{\frac{2}{3}} = (\sqrt[3]{27})^2 = (3)^2 = 3^2 = \boxed{9}$$

$$\text{II, VIII} \quad ⑱ \frac{b^3}{b^9} = b^{3-9} = b^{-6} = \boxed{\frac{1}{b^6}}$$

$$\text{III, V} \quad ⑲ (a^2 b^6)^3 = (a^2)^3 \cdot (b^6)^3 = a^{2 \cdot 3} b^{6 \cdot 3} = \boxed{a^6 b^{18}}$$

$$\text{I, X} \quad ⑳ 4^{\frac{3}{4}} \cdot 4^{\frac{1}{7}} = 4^{\frac{3}{4} + \frac{1}{7}} = 4^{\frac{21}{28} + \frac{4}{28}} = 4^{\frac{25}{28}} = \boxed{(\sqrt[28]{4})^{25}}$$

$$\text{V, X} \quad ㉑ (b^{\frac{1}{6}})^5 = b^{\frac{1}{6} \cdot 5} = b^{\frac{5}{6}} = \boxed{(\sqrt[6]{b})^5}$$

$$\text{I, VI} \quad ㉒ 7^{\frac{5}{8}} \cdot 7^{\frac{3}{8}} = 7^{\frac{5}{8} + \frac{3}{8}} = 7^{\frac{8}{8}} = 7^1 = \boxed{7}$$

$$\text{II, VIII} \quad ㉓ \frac{8^5}{8^7} = 8^{5-7} = 8^{-2} = \frac{1}{8^2} = \boxed{\frac{1}{64}}$$

$$\text{III, V} \quad ㉔ (x^4 y^8)^4 = (x^4)^4 \cdot (y^8)^4 = x^{4 \cdot 4} \cdot y^{8 \cdot 4} = \boxed{x^{16} y^{32}}$$

Problem Set D

$$\text{V, IX } (25) \left(x^{\frac{1}{2}}\right)^{\frac{1}{3}} = x^{\frac{1}{2} \cdot \frac{1}{3}} = x^{\frac{1}{6}} = \boxed{\sqrt[6]{x}}$$

$$\text{IV, V } (26) \left(\frac{b^{-2}}{c^{-3}}\right)^{-4} = \frac{(b^{-2})^{-4}}{(c^{-3})^{-4}} = \frac{b^{-2(-4)}}{c^{-3(-4)}} = \boxed{\frac{b^8}{c^{12}}}$$

$$\text{I, VI } (27) a^3 \cdot a^{-2} = a^{3+(-2)} = a^1 = \boxed{a}$$

$$\text{II, IX } (28) \frac{b^{\frac{5}{12}}}{b^{\frac{1}{6}}} = b^{\frac{5}{12} - \frac{1}{6}} = b^{\frac{5}{12} - \frac{2}{12}} = b^{\frac{3}{12}} = b^{\frac{1}{4}} = \boxed{\sqrt[4]{b}}$$

$$\text{IV, V } (29) \left(\frac{x^{-4}}{y^{-8}}\right)^{-2} = \frac{(x^{-4})^{-2}}{(y^{-8})^{-2}} = \frac{x^{-4(-2)}}{y^{-8(-2)}} = \boxed{\frac{x^8}{y^{16}}}$$

$$\text{I, VI } (30) 3^{-7} \cdot 3^8 = 3^{-7+8} = 3^1 = \boxed{3}$$

$$\text{V, IX } (31) \left(4^{\frac{1}{3}}\right)^{\frac{3}{2}} = 4^{\frac{1}{3} \cdot \frac{3}{2}} = 4^{\frac{1}{2}} = \sqrt{4} = \boxed{2}$$

$$\text{II, X } (32) \frac{x^{\frac{3}{4}}}{x^{\frac{1}{6}}} = x^{\frac{3}{4} - \frac{1}{6}} = x^{\frac{9}{12} - \frac{2}{12}} = x^{\frac{7}{12}} = \boxed{(\sqrt[12]{x})^7}$$

Problem Set E

$$\text{V, VI } (33) \left(y^{-\frac{3}{4}}\right)^{-\frac{4}{3}} = y^{-\frac{3}{4}(-\frac{4}{3})} = y^1 = \boxed{y}$$

$$\begin{aligned} \text{I, X } (34) 125^3 \cdot 125^{-\frac{7}{3}} &= 125^{3+(-\frac{7}{3})} = 125^{\frac{2}{3}} = (\sqrt[3]{125})^2 \\ &= (5)^2 \\ &= \boxed{25} \end{aligned}$$

$$\text{II, VI } (35) \frac{b^{-\frac{7}{9}}}{b^{-\frac{16}{9}}} = b^{-\frac{7}{9} - (-\frac{16}{9})} = b^{-\frac{7}{9} + \frac{16}{9}} = b^{\frac{9}{9}} = b^1 = \boxed{b}$$

$$\text{V, VIII } (36) (3^3)^{-1} = 3^{3(-1)} = 3^{-3} = \frac{1}{3^3} = \boxed{\frac{1}{27}}$$

$$\text{II, VI } (37) \frac{5^{\frac{7}{6}}}{5^{\frac{1}{6}}} = 5^{\frac{7}{6} - \frac{1}{6}} = 5^{\frac{6}{6}} = 5^1 = \boxed{5}$$

$$\text{I, X } (38) a^{-\frac{7}{5}} \cdot a^2 = a^{-\frac{7}{5} + 2} = a^{\frac{3}{5}} = \boxed{(\sqrt[5]{a})^3}$$

$$\text{V, VIII } (39) (7^{-1})^2 = 7^{-1(2)} = 7^{-2} = \frac{1}{7^2} = \boxed{\frac{1}{49}}$$

$$\text{V, VI } (40) \left(2^{\frac{4}{7}}\right)^{\frac{7}{4}} = 2^{\frac{4}{7} \cdot \frac{7}{4}} = 2^1 = \boxed{2}$$

Problem Set F

$$\text{V, VIII } \textcircled{41} (x^2)^{-2} = x^{2(-2)} = x^{-4} = \boxed{\frac{1}{x^4}}$$

$$\text{I, VI } \textcircled{42} y^{\frac{3}{7}} \cdot y^{\frac{4}{7}} = y^{\frac{3}{7} + \frac{4}{7}} = y^{\frac{7}{7}} = y^1 = \boxed{y}$$

$$\text{I, III } \textcircled{43} 8^{-\frac{2}{5}} \cdot 8^{\frac{2}{5}} = 8^{-\frac{2}{5} + \frac{2}{5}} = 8^{\frac{0}{5}} = 8^0 = \boxed{1}$$

$$\text{III, V } \textcircled{44} (x^6 y^2 z^3)^7 = (x^6)^7 (y^2)^7 (z^3)^7 = x^{6 \cdot 7} y^{2 \cdot 7} z^{3 \cdot 7} \\ = \boxed{x^{42} y^{14} z^{21}}$$

$$\text{VIII, III } \textcircled{45}^* (8x)^{-2} = \frac{1}{(8x)^2} = \frac{1}{8^2 x^2} = \boxed{\frac{1}{64x^2}}$$

$$\text{III, V } \textcircled{46} (a^2 b^3 c)^4 = (a^2)^4 (b^3)^4 c^4 = a^{2 \cdot 4} b^{3 \cdot 4} c^4 \\ = \boxed{a^8 b^{12} c^4}$$

$$\text{I, X } \textcircled{47} x^{\frac{1}{6}} \cdot x^{\frac{1}{2}} = x^{\frac{1}{6} + \frac{1}{2}} = x^{\frac{1}{6} + \frac{3}{6}} = x^{\frac{4}{6}} = x^{\frac{2}{3}} = \boxed{(\sqrt[3]{x})^2}$$

$$\text{V, VIII } \textcircled{48} (2^{-3})^2 = 2^{-3 \cdot 2} = 2^{-6} = \frac{1}{2^6} = \boxed{\frac{1}{64}}$$

$$\text{I, VII } \textcircled{49} c^{\frac{1}{4}} \cdot c^{-\frac{1}{4}} = c^{\frac{1}{4} + (-\frac{1}{4})} = c^0 = \boxed{1}$$

$$\text{VIII, IV } \textcircled{50}^* (3y)^{-3} = \frac{1}{(3y)^3} = \frac{1}{3^3 y^3} = \boxed{\frac{1}{27y^3}}$$

Three Rule Problems

Simplify each expression below and write the Roman numerals of the exponent rules that you used in the margin. Do not leave your answers with fraction exponents or negative exponents.

Problem Set G

$$\text{V, VIII, X } (51) (16^{-3})^{\frac{1}{4}} = 16^{-3 \cdot \frac{1}{4}} = 16^{-\frac{3}{4}} = \frac{1}{16^{\frac{3}{4}}} = \frac{1}{(\sqrt[4]{16})^3} = \frac{1}{2^3} = \boxed{\frac{1}{8}}$$

$$\text{I, III, V } (52) (a^3 \cdot a^6 \cdot y^2)^6 = (a^{3+6} y^2)^6 = (a^9 y^2)^6 = (a^9)^6 (y^2)^6 \\ = a^{9 \cdot 6} y^{2 \cdot 6} \\ = \boxed{a^{54} y^{12}}$$

$$\text{VI, I, V } (53) (x \cdot x^3)^4 = (x^1 \cdot x^3)^4 = (x^{1+3})^4 = (x^4)^4 \\ = x^{4 \cdot 4} \\ = \boxed{x^{16}}$$

$$\text{II, III, V } (54) \left(\frac{y^4 x^5}{x^3}\right)^7 = (y^4 x^{5-3})^7 = (y^4 x^2)^7 = (y^4)^7 (x^2)^7 \\ = y^{4 \cdot 7} x^{2 \cdot 7} \\ = \boxed{y^{28} x^{14}}$$

$$\text{I, III, V } (55) (x^4 \cdot b^2 \cdot b^3)^5 = (x^4 \cdot b^{2+3})^5 = (x^4 \cdot b^5)^5 \\ = (x^4)^5 (b^5)^5 \\ = x^{4 \cdot 5} b^{5 \cdot 5} \\ = \boxed{x^{20} b^{25}}$$

$$\text{VI, I, V } (56) (b \cdot b^8)^5 = (b^1 \cdot b^8)^5 = (b^{1+8})^5 = (b^9)^5 = b^{9 \cdot 5} \\ = \boxed{b^{45}}$$

$$\text{V, VIII, X } (57) (27^{-2})^{\frac{1}{3}} = 27^{-2 \cdot \frac{1}{3}} = 27^{-\frac{2}{3}} = \frac{1}{27^{\frac{2}{3}}} = \frac{1}{(\sqrt[3]{27})^2} \\ = \frac{1}{3^2} \\ = \boxed{\frac{1}{9}}$$

$$\text{II, IV, V } (58) \left(\frac{y^6}{x^2 y^2}\right)^{10} = \left(\frac{y^{6-2}}{x^2}\right)^{10} = \left(\frac{y^4}{x^2}\right)^{10} = \frac{(y^4)^{10}}{(x^2)^{10}} = \frac{y^{4 \cdot 10}}{x^{2 \cdot 10}} \\ = \boxed{\frac{y^{40}}{x^{20}}}$$

Problem Set H

$$\begin{aligned} \text{I, VII, IX } \textcircled{59} (a^{-5} \cdot b^5 \cdot a^5)^6 &= (a^{-5+5} \cdot b^5)^6 = (a^0 \cdot b^5)^6 \\ &= (1 \cdot b^5)^6 \\ &= (b^5)^6 \\ &= b^{5 \cdot 6} \\ &= \boxed{b^{30}} \end{aligned}$$

$$\begin{aligned} \text{I, II, IX } \textcircled{60} (a^7 \cdot a^{-6})^{\frac{1}{4}} &= (a^{7+(-6)})^{\frac{1}{4}} = (a^1)^{\frac{1}{4}} = a^{1 \cdot \frac{1}{4}} \\ &= a^{\frac{1}{4}} \\ &= \boxed{\sqrt[4]{a}} \end{aligned}$$

$$\begin{aligned} \text{IV, V, VI } \textcircled{61} \left(\frac{x^6}{y^6}\right)^{\frac{1}{6}} &= \frac{(x^6)^{\frac{1}{6}}}{(y^6)^{\frac{1}{6}}} = \frac{x^{6 \cdot \frac{1}{6}}}{y^{6 \cdot \frac{1}{6}}} = \frac{x^1}{y^1} = \boxed{\frac{x}{y}} \end{aligned}$$

$$\begin{aligned} \text{II, VII, IX } \textcircled{62} \left(\frac{bc^2}{c^2}\right)^{\frac{3}{8}} &= (bc^{2-2})^{\frac{3}{8}} = (bc^0)^{\frac{3}{8}} = (b \cdot 1)^{\frac{3}{8}} \\ &= (b)^{\frac{3}{8}} \\ &= b^{\frac{3}{8}} \\ &= \boxed{(\sqrt[8]{b})^3} \end{aligned}$$

$$\begin{aligned} \text{II, VII, IX } \textcircled{63} \left(\frac{a^5 b}{a^5}\right)^{\frac{2}{7}} &= (a^{5-5} b)^{\frac{2}{7}} = (a^0 b)^{\frac{2}{7}} = (1 \cdot b)^{\frac{2}{7}} \\ &= (b)^{\frac{2}{7}} \\ &= b^{\frac{2}{7}} \\ &= \boxed{(\sqrt[7]{b})^2} \end{aligned}$$

$$\begin{aligned} \text{I, V, IX } \textcircled{64} (3^{-7} \cdot 3^8)^{\frac{1}{5}} &= (3^{-7+8})^{\frac{1}{5}} = (3^1)^{\frac{1}{5}} = 3^{1 \cdot \frac{1}{5}} = 3^{\frac{1}{5}} \\ &= \boxed{\sqrt[5]{3}} \end{aligned}$$

$$\begin{aligned} \text{I, VII, IX } \textcircled{65} (x^4 \cdot y^2 \cdot x^{-4})^2 &= (x^{4+(-4)} y^2)^2 = (x^0 y^2)^2 \\ &= (1 \cdot y^2)^2 \\ &= (y^2)^2 \\ &= y^{2 \cdot 2} \\ &= \boxed{y^4} \end{aligned}$$

$$\begin{aligned} \text{IV, V, VI } \textcircled{66} \left(\frac{a^{\frac{1}{4}}}{b^{\frac{1}{4}}}\right)^4 &= \frac{(a^{\frac{1}{4}})^4}{(b^{\frac{1}{4}})^4} = \frac{a^{\frac{1}{4} \cdot 4}}{b^{\frac{1}{4} \cdot 4}} = \frac{a^1}{b^1} = \boxed{\frac{a}{b}} \end{aligned}$$

Problem Set I

$$\text{II, III, V } \textcircled{67} \left(\frac{x^7 y^8}{y^5 x^3} \right)^3 = (x^{7-3} \cdot y^{8-5})^3 = (x^4 \cdot y^3)^3 \\ = (x^4)^3 (y^3)^3 \\ = x^{4 \cdot 3} y^{3 \cdot 3} \\ = \boxed{x^{12} y^9}$$

$$\text{VIII, III, V } \textcircled{68} (2y^5)^{-3} = \frac{1}{(2y^5)^3} = \frac{1}{2^3 (y^5)^3} = \frac{1}{8y^{5 \cdot 3}} \\ = \boxed{\frac{1}{8y^{15}}}$$

$$\text{II, I, VIII } \textcircled{69} \left(\frac{b^5}{b^4} \right)^{-4} = (b^{5-4})^{-4} = (b^1)^{-4} = b^{1(-4)} \\ = b^{-4} \\ = \boxed{\frac{1}{b^4}}$$

$$\text{II, III, V } \textcircled{70} \left(\frac{a^9 b^5}{b^2 a^3} \right)^4 = (a^{9-3} \cdot b^{5-2})^4 = (a^6 b^3)^4 = (a^6)^4 (b^3)^4 \\ = a^{6 \cdot 4} b^{3 \cdot 4} \\ = \boxed{a^{24} b^{12}}$$

$$\text{VIII, III, V } \textcircled{71} (5x^4)^{-2} = \frac{1}{(5x^4)^2} = \frac{1}{5^2 (x^4)^2} = \frac{1}{25x^{4 \cdot 2}} = \boxed{\frac{1}{25x^8}}$$

$$\text{II, I, VIII } \textcircled{72} \left(\frac{x^{-5}}{x^{-6}} \right)^{-2} = (x^{-5-(-6)})^{-2} = (x^{-5+6})^{-2} \\ = (x^1)^{-2} \\ = x^{1(-2)} \\ = x^{-2} \\ = \boxed{\frac{1}{x^2}}$$

Using the Variation of Rule X

Simplify each expression below and write the Roman numerals of the exponent rules that you used in the margin.

Problem Set J

$$\text{II, X'} \quad (73) \sqrt[3]{\frac{8^7}{8^3}} = \sqrt[3]{8^{7-3}} = \sqrt[3]{8^4} = (\sqrt[3]{8})^4 = 2^4 = 16$$

$$\text{I, X'} \quad (74) \sqrt[4]{(81^{\frac{1}{3}})^6} = \sqrt[4]{81^{\frac{1}{3} \cdot 6}} = \sqrt[4]{81^2} = (\sqrt[4]{81})^2 = 3^2 = 9$$

$$\text{VI, I, X'} \quad (75) \sqrt{9 \cdot 9^2} = \sqrt{9^1 \cdot 9^2} = \sqrt{9^{1+2}} = \sqrt{9^3} = (\sqrt{9})^3 = 3^3 = 27$$

$$\text{I, X'} \quad (76) \sqrt[3]{(27^{\frac{1}{2}})^4} = \sqrt[3]{27^{\frac{1}{2} \cdot 4}} = \sqrt[3]{27^2} = (\sqrt[3]{27})^2 = 3^2 = 9$$

$$\text{II, X'} \quad (77) \sqrt{\frac{25^8}{25^5}} = \sqrt{25^{8-5}} = \sqrt{25^3} = (\sqrt{25})^3 = 5^3 = 125$$

$$\text{VI, I, X'} \quad (78) \sqrt[4]{16^4 \cdot 16} = \sqrt[4]{16^4 \cdot 16^1} = \sqrt[4]{16^{4+1}} = \sqrt[4]{16^5} = (\sqrt[4]{16})^5 = 2^5 = 32$$

$$\text{II, X'} \quad (79) \sqrt{(16^{-12})^{-\frac{1}{4}}} = \sqrt{16^{-12(-\frac{1}{4})}} = \sqrt{16^3} = (\sqrt{16})^3 = 4^3 = 64$$

$$\text{II, X'} \quad (80) \sqrt[4]{\frac{81^{-3}}{81^{-5}}} = \sqrt[4]{81^{-3-(-5)}} = \sqrt[4]{81^{-3+5}} = \sqrt[4]{81^2} = (\sqrt[4]{81})^2 = 3^2 = 9$$

$$\text{I, X'} \quad (81) \sqrt[3]{27^2 \cdot 27^2} = \sqrt[3]{27^{2+2}} = \sqrt[3]{27^4} = (\sqrt[3]{27})^4 = 3^4 = 81$$